

1. In a single-parity-check code, a single parity bit b_1 is appended to a block of k information bits, $m_1, m_2, \dots, m_k$. The parity bit is chosen so that the codeword satisfies the even parity rule:

$$m_1 + m_2 + \dots + m_k + b_1 = 0, \quad \text{mod } 2$$

- a) For $k = 3$, construct the 2^k valid codewords in the code defined by this rule.
 b) Show that the minimum Hamming distance among valid codewords is $d_{\min} = 2$.

a) valid codewords:

m_1	m_2	m_3	b_1
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

b) For every valid codeword, it contains even numbers of "1"

Hence, $d_{\min} = "0011" - "0000" = 2$

2. Consider a linear block code with total length $n = 13$ bits. If the code tolerates up to 3 bit errors, at least how many parity check bits is needed?
 (Hint: You need not construct the specific code)

$$e=3, \quad 2^r \geq \sum_{m=0}^e \binom{n}{m} = \sum_{m=0}^3 \binom{13}{m} = 1 + 13 + 78 + 286 = 378$$

$$\Leftrightarrow r \geq \log_2 378 \Rightarrow r_{\min} = \lceil \log_2 378 \rceil = 9$$

3. Consider a (7,4) Hamming code with generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

- a) Determine the parity check matrix H .
 b) Decode the message if the received codeword is 0101011.

a) $G = [I_4 : Q] \Leftrightarrow Q = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$

$$H^T = \begin{pmatrix} Q \\ I_r \end{pmatrix} \Leftrightarrow H = [Q^T : I_r] = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}$$

b) $b = 0101011$
 $bH^T = (0101011) \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = (110) + (101) + (010) + (001) = 000$
 $\Rightarrow$ received codeword is a valid codeword
 $\Rightarrow$ message = 0101

4. Consider a (7,3) linear block code with generator matrix

$$G = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

- a) Construct the generator matrix $\bar{G}$ of an equivalent systematic code and brief on why they are equivalent. (Hint: Conduct row transformations on G)
 b) Determine the minimum Hamming distance $d_{\min}$. What is the error correction capacity t and the error detection capacity e ?
 c) Can you decode the message if the received codeword is $r_1 = 0100010$ or $r_2 = 1111001$?

a)

$$G = \begin{pmatrix} g_1 \\ g_2 \\ g_3 \end{pmatrix}$$

$$A = \left\{ a = (a_1, a_2, \dots, a_6) \mid a = xG = (x_1, x_2, x_3) \begin{pmatrix} g_1 \\ g_2 \\ g_3 \end{pmatrix} = x_1 g_1 \oplus x_2 g_2 \oplus x_3 g_3, \quad x_1, x_2, x_3 \in \{0, 1\} \right\}$$

$$B = \left\{ a' = (a'_1, a'_2, \dots, a'_6) \mid a' = (x_1, x_2, x_3) E \begin{pmatrix} g_1 \\ g_2 \\ g_3 \end{pmatrix}, \quad E \text{ is row transform matrix} \right\}$$

Obviously, $A = B \Leftrightarrow \bar{G} = EG$. Let $\bar{G} = \begin{pmatrix} g_1 + g_3 \\ g_1 + g_2 \\ g_1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$

b) valid codewords $d_{\min} = 4$
 $e = t = 1$

0000000
0010111
0101110
0111001
1001011
1011100
1100101
1110010

c) $r_1 = 0100010$, $|"0100010" - "0000000"| = |"0100010" - "1110010"| = 2 = t_{\min}$
 $\Rightarrow$ correct codewords may be "0000000" or "1110010"
 $\Leftrightarrow$ message may be "000" or "001"
 $\Rightarrow$ can't be decoded

$r_2 = "1111001"$, $|"1111001" - "0111001"| = 1$ is the minimum distance
 $\Rightarrow$ correct codeword is "0111001"
 $\Leftrightarrow$ message is "011" $\Rightarrow$ can be decoded